Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 3.1 – Direct calculations

In an orthonormal basis of the plane, consider the vectors $\vec{u} = (1; -2)$ and $\vec{v} = (4; 3)$.

1. Calculate the scalar product $\vec{u} \cdot \vec{v}$, then the norms $\|\vec{u}\|$ and $\|\vec{v}\|$.
2. Calculate the coordinates of $\vec{u} + \vec{v}$ and deduce $\|\vec{u} + \vec{v}\|^2$.
3. Recover this last result using the identity involving $\vec{u} \cdot \vec{v}$.

Exercise 3.2 – A right triangle?

The plane is equipped with an orthonormal frame. Given the points $A(1; 1)$, $B(4; 2)$ and $C(0; 4)$.

1. Calculate the coordinates of $\overrightarrow{AB}$ and $\overrightarrow{AC}$, then their scalar product.
2. Deduce that the triangle ABC is right-angled; state at which vertex.
3. Calculate the lengths of the two perpendicular sides, then the area of the triangle in area units.

Exercise 3.3 – Length without a frame

The quadrilateral $ABCD$ is a parallelogram with $AB = 4$, $AD = 3$ and $\widehat{BAD} = 60°$. The lengths are expressed in the same unit.

1. Draw a figure and calculate the scalar product $\overrightarrow{AB} \cdot \overrightarrow{AD}$.
2. Determine the exact lengths of the diagonals $[AC]$ and $[BD]$. Explain how the vectors corresponding to the diagonals are expressed in terms of $\overrightarrow{AB}$ and $\overrightarrow{AD}$.
3. State which diagonal is longer and check that this result is consistent with your figure.

Exercise 3.4 – Recovering an angle

Two nonzero vectors $\vec{u}$ and $\vec{v}$ satisfy $\|\vec{u}\| = 2$, $\|\vec{v}\| = 6$ and $\vec{u} \cdot \vec{v} = -6$. Let θ be their unoriented angle, between $0°$ and $180°$.

1. Determine the exact value of $\cos\theta$, then the measure of θ.

2. Explain how the sign of the scalar product made it possible to predict that the angle would be obtuse.

Exercise 3.5 – A variable locus

The points A and B are distinct and fixed. Let I be the midpoint of $[AB]$. For a fixed real number k, we look for the set of points M of the plane that satisfy $\overrightarrow{MA} \cdot \overrightarrow{MB} = k$.

1. Express this scalar product in terms of MI and AB using Chasles's relation and a scalar identity.
2. Describe precisely the set sought according to the value of k. For each case, state whether it is empty, a single point or a circle; then give its characteristics.
3. Explain the role of the fact that a squared distance is positive or zero.

Exercise 3.6 – Work done by a force

An object moves in a straight line from A to B over a distance of 4 m. It is subject to a constant force $\vec{F}$ of magnitude 50 N. The work done by this force is defined by $W = \vec{F} \cdot \overrightarrow{AB}$ and is expressed in joules.

1. Calculate W when the angle between $\vec{F}$ and $\overrightarrow{AB}$ is $60°$.
2. Repeat the calculation for an angle of $120°$.
3. Compare the signs of the two works and explain, in each situation, whether the force helps the displacement or opposes it.

Exercise 3.7 – Contrapositive

We wish to prove the following statement, valid for two vectors of the plane:

$$\|\vec{u}+\vec{v}\|^2 \neq \|\vec{u}\|^2 + \|\vec{v}\|^2 \quad \Longrightarrow \quad \vec{u} \text{ and } \vec{v} \text{ are not orthogonal.}$$

1. State the contrapositive of this implication.
2. Prove this contrapositive using a scalar identity, then conclude about the initial statement.

Exercise 3.8 – Delayed review

A triangle ABC satisfies $AB = 3$, $AC = 4$ and $BC = 5$. We look for $\overrightarrow{AB} \cdot \overrightarrow{AC}$ without using coordinates.

1. Calculate this scalar product using a formula involving the three lengths.
2. Recover the result by determining the nature of the triangle. Name the theorem used and its numerical hypothesis.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 9 p. 250, questions a and b. Calculate the scalar product in the orthonormal basis and detail the two products of coordinates.

CQFD, exercise 11 p. 250, questions 1, 2 and 3. Use in turn the coordinates, Chasles's relation and the fact that I is the midpoint.